Exercise 9.1: The OOD Problem Your CARLA model was trained on sunny daytime images. At deployment, it may encounter images from fog or night - conditions outside its ODD.

1. Why can a standard classifier not be trusted to signal when it receives an OOD input?

- A standard classifier is trained only to classify known training data.
- It is not trained to recognize unfamiliar inputs.
- Therefore, it can produce high-confidence predictions even for OOD images.
- The model cannot reliably say "I don't know."

Example

A model trained only on sunny images may classify a night image as "No Pedestrian" with 98% confidence even though it has never seen night conditions.

2. Why is silent failure (confidently wrong prediction) worse than uncertain failure in a safety-critical system?

- Silent failure is dangerous because the system is confidently wrong.
- The system has no warning that the prediction may be incorrect.
- Uncertain failure is safer because the system knows it is unsure and can take precautionary actions.
- In safety-critical systems, detecting uncertainty helps prevent accidents.

Example

A self-driving car confidently misses a pedestrian in a night image and fails to brake, whereas an uncertain prediction could trigger a safety response.

Exercise 9.2: Baseline OOD Detection Describe the maximum softmax probability (MSP) baseline for OOD detection. How does it work, and what are its main limitations?

- Maximum Softmax Probability (MSP) is a simple baseline method for OOD detection.
- It uses the highest predicted probability from the classifier as the confidence score.
- Images with high confidence are considered in-distribution, while images with low confidence are considered OOD.
- The method is easy to implement and does not require modifying the model.

Limitation:

- Neural networks can produce high-confidence predictions for OOD inputs.
- Therefore, MSP may fail to detect some OOD samples and can result in confident but incorrect predictions.

Example: A model trained on sunny images may classify a night image with 99% confidence even though it has never seen nighttime conditions.

MSP assumes that unfamiliar inputs produce low confidence, but neural networks can still be confidently wrong.

Exercise 9.3: Alternative Methods Describe one more advanced OOD detection method (e.g., energy score or Mahalanobis distance). How does it improve over MSP?

- Mahalanobis Distance is a feature-based OOD detection method.
- It uses deep features extracted from the neural network instead of prediction confidence.
- The distance between a test image and the distribution of training features is computed.
- Images with large distances are considered OOD.

Improvement over MSP:

- MSP relies only on prediction confidence.
- Mahalanobis uses information from the learned feature space.
- It can detect OOD inputs even when the classifier is highly confident.
- Therefore, it is often more reliable than MSP for OOD detection.

Example:

A night image may receive a high-confidence prediction from the classifier, but its feature representation can still be far from the training distribution, allowing Mahalanobis distance to identify it as OOD.

Exercise 9.4: Visualising the Distribution Shift

1. **Display five in-distribution (sunny / daytime) images next to five fog and five night images.**

Sunny Images

Fog Images

Night Images

2. Also display five images from the different CARLA town. How do they differ from your training images, and how does that difference compare to the fog / night shift?

Different Town Images

- The fog images have reduced visibility and contrast compared to the training images.
- Night images differ significantly due to low illumination and strong lighting effects.
- The different-town images retain the same daytime and weather conditions as the training data but contain different buildings, road layouts, and vegetation.
- Therefore, the fog and night shifts are stronger distribution shifts than the different-town images.

3. Compute the mean softmax confidence of each of your three models on in-distribution vs. fog/night images. Is the model more or less confident on these inputs?

Model	Sunny	Fog	Night
Pedestrian	0.904	0.990	0.972
Vehicle	0.910	0.820	0.764
Traffic Light	0.959	0.822	0.885

Pedestrian Model

- Confidence increased on fog and night images.
- The model remained highly confident even on OOD data.
- This suggests possible overconfidence and highlights a limitation of MSP.

Vehicle Model

- Confidence decreased in fog and night conditions.
- The largest drop occurred at night.
- The model became less certain when encountering unfamiliar conditions.

Traffic Light Model

- Confidence decreased in fog and night images.
- The reduction was smaller than for the vehicle model.
- The model was still reasonably confident at night.